

# synaptic Technical Reference

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## synaptic project

*Open-source LaTeX community*

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**Abstract:** This reference describes synaptic’s module graph, load-time invariants, namespace rules, font and colour subsystems, extension points, generated-file policy, and automated verification workflow.

**Keywords:** LaTeX3, architecture, l3build, regression testing

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## 1 Architecture

`synaptic.sty`

```

`-- synaptic-base.sty
    |-- synaptic-lang-{en,zh}.def
    |-- synaptic-color.sty
    |-- synaptic-fonts.sty
    |-- synaptic-layout.sty
    |-- synaptic-theorem.sty
    |-- synaptic-title.sty      shared metadata/renderers
`-- mode module(s)
    |-- synaptic-journal.sty    journal
    |-- synaptic-book.sty      book
    |-- synaptic-lecture.sty    lecture
    |-- synaptic-boxes.sty      lecture
    |-- synaptic-notes.sty      notes

```

The thin shell forwards user options to `synaptic-base` and defines a small set of collision-resistant convenience commands. The base module checks the engine and class, parses options, loads labels, validates book mode, and dispatches core and mode-specific modules.

| Mode    | Class invariant                            | Extra modules  |
|---------|--------------------------------------------|----------------|
| journal | KOMA article/report                        | journal        |
| book    | chapter-based KOMA class with front matter | book           |
| lecture | KOMA article/report                        | lecture, boxes |
| notes   | KOMA article/report                        | notes          |

## 2 Configuration lifecycle

The package-level `synaptic` key family is evaluated before module loading. Mode, language, font set, title layout, measure, density, numbering, delivery colour mode, extra math alphabets, and fine-breaking behaviour therefore become structural invariants. The runtime key family exposes only the state that can be applied without reloading packages: `theme`, `toc`, and `toc-layout`. Frozen or unknown keys produce diagnostic warnings.

Book mode is rejected unless both chapter and front-matter interfaces exist. LuaLaTeX and KOMA-Script requirements are fatal diagnostics rather than delayed undefined-control-sequence failures.

### 3 Subsystems

#### 3.1 Colour tokens

Built-in themes are stored in a global property list. Activating a theme maps its primary, secondary, and accent colours to semantic public names and derives surface, border, rule, link, readable-muted, decorative-muted, and text tokens. Print and mono delivery modes alter derived roles without changing component code. Danger, warning, and success colours remain semantic constants so their meaning does not change with brand colour.

```
\SynapticDefineTheme{name}{primary}{secondary}[accent]
\SynapticUseTheme{name}
```

The arguments are xcolor expressions. Defining a custom theme registers and activates it.

#### 3.2 Font selection

Font discovery is implemented entirely with `\IfFontExistsTF`. Each candidate is a complete roman/sans/mono/math bundle; auto selection never picks a family whose companions are missing. The fixed priority is XCharter, Libertinus, STIX Two, then Latin Modern. An explicit incomplete choice emits a package error with the missing bundle requirements.

For `lang=zh`, ctex is loaded without a preset font scheme. Noto CJK SC is preferred and Fandol is the TeX Live fallback. Synthetic weight/slant is limited to faces for which the preferred family has no native counterpart.

#### 3.3 Layout and document structure

Geometry combines mode defaults with explicit measure and density profiles. Journal and lecture favour moderate line lengths; notes supports compact single/two-column composition; book uses mirrored inner/outer margins, binding correction, outer running heads, and empty duplex blanks. Widow and club penalties, vertical rhythm, list spacing, line spacing, and paragraph indentation are explicit rather than hidden behind global bad-box suppression.

KOMA heading fonts and skips are installed in the `begindocument/before` hook. Hyperref, bookmark, microtype, captions, lists, caption punctuation, and mode-aware equation numbering are configured centrally.

#### 3.4 Titles and metadata

Title values are held in expl3 token lists and sequences. Authors and their affiliation marks use parallel sequences; affiliations and emails are repeatable. A common metadata model feeds four renderers: inline journal, book title sequence, course

masthead, and compact notes masthead. The title command validates a non-empty title and controls optional TOC placement without silently resetting the user’s page numbering. Standard LaTeX metadata is an input adapter, not a redefinition of `\maketitle`.

PDF information is passed as source tokens to hyperref. Hyperref performs the Unicode PDF-string conversion once; pre-converting and then calling `\hypersetup` would double-encode metadata.

### 3.5 Theorems and boxes

thmtools defines plain, definition, and remark styles on top of amsthm. The built-in theorem family uses a mode-aware shared counter: chapter scope for books and section scope elsewhere by default. Lecture mode wraps core theorem environments with tcolorbox; examples and exercises can share that counter, while warning and remark cards remain unnumbered by default. Public short names use the `syn` prefix; v2.0 generic aliases are conditional compatibility shims.

## 4 Namespace policy

| Scope              | Pattern                            | Example                           |
|--------------------|------------------------------------|-----------------------------------|
| Public command     | <code>\Synaptic...</code>          | <code>\SynapticDefineTheme</code> |
| Public environment | <code>syn...</code>                | <code>synwarning</code>           |
| Internal function  | <code>\__synaptic_...:</code>      | title rendering helper            |
| Module interface   | <code>\synaptic_...:</code>        | mode accessor                     |
| Local state        | <code>\l_synaptic_...</code>       | selected theme token list         |
| Global registry    | <code>\g_synaptic_...</code>       | theme property list               |
| Label constant     | <code>\c_synaptic_label_...</code> | theorem label                     |

New public names should avoid common nouns. Compatibility aliases must use `\ProvideDocumentCommand` or an explicit existence check.

## 5 Extension points

- Add labels in a new `synaptic-lang-*.def` file, then extend the language key and definition-file dispatch.
- Add a complete font predicate, setter, explicit dispatch branch, and a deliberate location in the auto-selection chain.
- Add modes in the base key family, layout dispatch, and mode-module loader. State the required KOMA class capability and test it.
- Build visual extensions from semantic colour tokens rather than fixed RGB values.

## 6 Source and release workflow

`synaptic.dtx` is the source of truth. DocStrip uses `synaptic.ins` to generate the installable `.sty` and `.def` files. The generated copies under `tex/latex/synaptic/` are committed for zero-install use and must remain byte-identical to fresh extraction.

```
l3build unpack
l3build check
l3build doc
l3build ctan
```

The nine-group regression suite covers journal titles and statements, book matter and chapter numbering, lecture navigation and boxes, notes cards and standard metadata, structural options, custom themes, and Chinese typesetting. CI also builds the documented source, four manuals, and all examples.

## 7 Compatibility notes

Version 2.2 retains lower-case book matter helpers and generic lecture box names as conditional v2.0 aliases. The former `theme` key remains the palette selection interface. New documents should use the namespaced public forms. Removed `style=` and `color=` options remain explicit errors with migration hints.